POISSON LIMITS FOR LARGE BOXES IN SPARSE RANDOM ALLOCATIONS

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ABSTRACT. Let $(Y_1,\dots,Y_n)$ be the random allocation of m balls into n boxes in which an allocation $(y_1,\dots,y_n)$ has probability proportional to $\prod_i w_{y_i}$, and let $\xi_1,\dots,\xi_n$ be independent with $\mathbb{P}(\xi_i=k)$ proportional to $w_k\tau^k$, where τ is chosen so that $\mathbb{E}\xi_i=m/n$. Janson showed that when m/n converges to a positive limit below a threshold determined by the weights, the numbers of boxes with many balls are asymptotically Poisson and the largest boxes of the allocation are close in distribution to the largest of $\xi_1,\dots,\xi_n$, and he asked whether the same holds when $m/n\to 0$. Let a be the least $k\ge 1$ with $w_k>0$. The answer is positive for every sequence with $m\to\infty$ if every k with $w_k>0$ is a multiple of a, for instance if $w_1>0$, and it is negative along every sequence on which m is bounded. If some k with $w_k>0$ is not a multiple of a, there are sequences with $m\to\infty$ and $m/n\to 0$ along which the answer is negative. When the least such k, say b, is coprime to a, we characterize the sequences along which the answer is positive in terms of $n\mathbb{P}(\xi_1=b)$ and the residue of m modulo a; when $n\mathbb{P}(\xi_1=b)$ has a positive finite limit, the number of boxes with at least b balls converges to a Poisson distribution conditioned on a residue class. The proofs rest on an exact formula for the joint law of the boxes whose sizes are neither 0 nor a and on a local limit theorem for the binomial distribution with explicit constants.

1. INTRODUCTION

Let $w=(w_k)_{k\ge 0}$ be a sequence of nonnegative weights with $w_0>0$ and $w_k>0$ for some $k\ge 1$. For integers $m\ge 0$ and $n\ge 1$ let $\mathcal{B}_{m,n}$ be the set of $y=(y_1,\dots,y_n)\in\mathbb{Z}_{\ge 0}^n$ with $y_1+\cdots+y_n=m$, let $w(y)=\prod_{i=1}^n w_{y_i}$, and let $Z(m,n)=\sum_{y\in\mathcal{B}_{m,n}} w(y)$. When $Z(m,n)>0$, the random allocation $B_{m,n}=(Y_1,\dots,Y_n)$ is the random element of $\mathcal{B}_{m,n}$ with

$$\mathbb{P}(B_{m,n}=y)=\frac{w(y)}{Z(m,n)},\qquad y\in\mathcal{B}_{m,n}.$$

We think of Y_i as the number of balls in box i, write $\mathbf{Y}=(Y_1,\dots,Y_n)$, and write $Y_{(1)}\ge Y_{(2)}\ge\cdots\ge Y_{(n)}$ for $Y_1,\dots,Y_n$ in decreasing order. We follow the notation of Janson's survey [2], which treats random allocations together with simply generated random trees.

Let $\Phi(t)=\sum_k w_k t^k$, let $\rho\in[0,\infty]$ be its radius of convergence, and for $0\le t<\rho$ let $\Psi(t)=t\Phi'(t)/\Phi(t)$. Since $t\Psi'(t)$ is the variance of the probability distribution $(w_kt^k/\Phi(t))_{k\ge 0}$, the function Ψ is continuous and strictly increasing on $[0,\rho)$, and $\Psi(0)=0$. Let $\nu=\lim_{t\uparrow\rho}\Psi(t)\in(0,\infty]$ if $\rho>0$, and $\nu=0$ if $\rho=0$. When $m/n<\nu$, let $\tau_n\in[0,\rho)$ be the solution of $\Psi(\tau_n)=m/n$ and let $\xi^{(n)}$ be a random variable with

$$\mathbb{P}(\xi^{(n)}=k)=\frac{w_k\tau_n^k}{\Phi(\tau_n)},\qquad k\ge 0,$$

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so that $\mathbb{E}\xi^{(n)} = \Psi(\tau_n) = m/n$. Let $\boldsymbol{\xi} = (\xi_1, \dots, \xi_n)$ consist of independent copies of $\xi^{(n)}$, let $\xi_{(1)} \geq \cdots \geq \xi_{(n)}$ be the ξ_i in decreasing order, and let $S_n = \xi_1 + \cdots + \xi_n$. For $A \subseteq \mathbb{Z}_{\geq 0}$ and $x \in \mathbb{Z}_{\geq 0}^n$ let $N_A(x) = |\{i \leq n : x_i \in A\}|$, and write $N_A = N_A(\mathbf{Y})$ and $N_A^{\xi} = N_A(\boldsymbol{\xi})$; thus $N_{[h,\infty)}^{\xi}$ has the binomial distribution $\mathrm{Bin}(n, \mathbb{P}(\xi^{(n)} \geq h))$.

We use the total variation distance d_{TV}, the Kolmogorov distance $d_{\mathrm{K}}(X, X') = \sup_x |\mathbb{P}(X \leq x) - \mathbb{P}(X' \leq x)|$ and, for random variables with values in $\mathbb{Z}_{\geq 0} \cup \{\infty\}$, the distance

$$\tilde{d}_{\mathrm{K}}(X, X') = \sup_{x \in \mathbb{Z}_{\geq 0}} \frac{|\mathbb{P}(X \leq x) - \mathbb{P}(X' \leq x)|}{1 + x}$$

of [2, (19.13)]. Then $\tilde{d}_{\mathrm{K}} \leq d_{\mathrm{K}} \leq d_{\mathrm{TV}}$, and $\tilde{d}_{\mathrm{K}}(X_n, X) \to 0$ if and only if $X_n \to X$ in distribution [2, Lemma 19.5]. Here $\mathrm{Po}(\infty)$ denotes the distribution concentrated at ∞.

Suppose that $\nu > 0$, that $w_k > 0$ for infinitely many k, and that $m/n \to \lambda$ with $0 < \lambda < \nu$. Janson [2, Theorem 19.7] proved that then the following hold, where $h = h(n)$ are integers and $j \geq 1$ is fixed.

(i) If $n\mathbb{P}(\xi^{(n)} \geq h) \to \alpha \in [0, \infty]$, possibly for n in a subsequence, then $N_{[h,\infty)} \to \mathrm{Po}(\alpha)$ in distribution.
(ii) If $n\mathbb{P}(\xi^{(n)} \geq h) \to 0$, then $Y_{(1)} < h$ with probability tending to one.
(iii) If $n\mathbb{P}(\xi^{(n)} \geq h) \to \infty$, then $Y_{(j)} \geq h$ with probability tending to one.
(iv) $\tilde{d}_{\mathrm{K}}(N_{[h,\infty)}, N_{[h,\infty)}^{\xi}) \to 0$.
(v) $d_{\mathrm{K}}(Y_{(j)}, \xi_{(j)}) \to 0$.
(vi) $d_{\mathrm{TV}}(Y_{(1)}, \xi_{(1)}) \to 0$.

Theorem 19.7 of [2] is stated with a fixed random variable in place of $\xi^{(n)}$ when $m = \lambda n + o(\sqrt{n})$, and in the form above for every sequence with $m/n \to \lambda \in (0, \nu)$; by [2, Remark 19.8] it suffices that $c \leq m/n \leq C$ for constants $0 < c < C < \nu$. Janson asked about the remaining case.[1]

Problem (Janson [2, Problem 19.9])**.** *Is Theorem 19.7 (in the $\xi^{(n)}$ version) true also for $\lambda = 0 < \nu$?*

We answer this question. From now on $\nu > 0$, $n \to \infty$, $m = m(n) \geq 1$ and $m/n \to 0$, and we consider only n with $Z(m, n) > 0$; we do not assume that $w_k > 0$ for infinitely many k. (If $m = 0$, then $Y_i = \xi_i = 0$ for all i and (i)–(vi) hold trivially.) We say that the conclusions of Theorem 19.7 hold if (i)–(vi) hold for all integers $h(n)$ and every fixed j. Let

$$a = \min\{k \geq 1 : w_k > 0\}.$$

Every box holds either no ball or at least a balls.

Theorem 1.1. *Suppose that every k with $w_k > 0$ is a multiple of a; for instance, $w_1 > 0$. Then the conclusions of Theorem 19.7 hold if and only if $m \to \infty$. Moreover, if $m \to \infty$, then*

$$d_{\mathrm{TV}}\big((Y_{(1)}, \dots, Y_{(j)}), (\xi_{(1)}, \dots, \xi_{(j)})\big) \to 0$$

for every fixed j, and the total variation distance between the vectors $(Y_i \mathbf{1}\{Y_i \notin \{0, a\}\})_{i \leq n}$ and $(\xi_i \mathbf{1}\{\xi_i \notin \{0, a\}\})_{i \leq n}$ tends to zero.

Together with [2, Theorem 19.7 and Remark 19.8], Theorem 1.1 shows that if $w_1 > 0$ and $w_k > 0$ for infinitely many k, the conclusions of Theorem 19.7 hold whenever $m \to \infty$ and $\limsup m/n < \nu$: every subsequence has a further subsequence along which m/n converges to a limit in $[0, \nu)$, and each of the two theorems covers one kind of limit.

[1] In the arXiv version of [2] it is Problem 18.9.

If some k with $w_k > 0$ is not a multiple of a, then $a \geq 2$. We then let b be the least such k, so that $b > a$, and let $c_n = n\mathbb{P}(\xi^{(n)} = b)$. Since $\Phi(t) = w_0(1+O(t))$ and $\Psi(t) = (aw_a/w_0)t^a(1+O(t))$ as $t \to 0$, and $\tau_n \to 0$,

$$c_n = \frac{w_b}{w_0}\Big(\frac{w_0}{aw_a}\Big)^{b/a} m^{b/a} n^{1-b/a}\big(1 + O(\tau_n)\big). \tag{1.1}$$

Here and below, $O(\cdot)$ and $o(\cdot)$ refer to $n \to \infty$, with constants depending only on w.

Proposition 1.2.

(a) Suppose that $m = m_0$ for all n in an infinite set $\mathcal{N}$. Then, as $n \to \infty$ in $\mathcal{N}$, $n\mathbb{P}(\xi^{(n)} \geq 1) \to m_0/a$ and $\mathbb{P}(\xi_{(1)} = 0) \to e^{-m_0/a}$, while $Y_{(1)} \geq 1$ and $N_{[1,\infty)} \leq m_0/a$. Hence (i) and (iv) fail for $h = 1$, and (v) and (vi) fail for $j = 1$.

(b) Suppose that b exists. If $a \nmid m$, then $Y_{(1)} \geq b$, while $n\mathbb{P}(\xi^{(n)} \geq b) = c_n(1 + O(\tau_n))$. Hence, along every sequence with $a \nmid m$ and $c_n \to 0$, $d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)}) \to 1$, (i), (ii) and (iv) fail for $h = b$, and (v) and (vi) fail for $j = 1$. The sequence $m = b + a\lfloor n^{(b-a)/(2b)} \rfloor$ has $m \to \infty$, $m/n \to 0$, $a \nmid m$ and $c_n \asymp n^{-(b-a)/(2a)}$.

By Theorem 1.1 and Proposition 1.2, the conclusions of Theorem 19.7 hold along every sequence with $m \to \infty$ and $m/n \to 0$ if and only if every k with $w_k > 0$ is a multiple of a. For weights of span one, that is, when the k with $w_k > 0$ have greatest common divisor 1, this condition says that $w_1 > 0$. When b exists and is coprime to a, the answer depends on c_n and on the residue of m modulo a. We write X_c for a random variable with distribution $\mathrm{Po}(c)$.

Theorem 1.3. *Suppose that b exists and $\gcd(a, b) = 1$.*

(a) If $c_n \to \infty$, then the conclusions of Theorem 19.7 hold, and for every fixed j

$$d_{\mathrm{TV}}\big((Y_{(1)}, \dots, Y_{(j)}), (\xi_{(1)}, \dots, \xi_{(j)})\big) \to 0.$$

(b) If $c_n \to c \in (0, \infty)$ and $m \equiv \ell \pmod a$ for all n, with $\ell \in \{0, \dots, a-1\}$, then $N_{[b,\infty)}$ converges in total variation to the conditional distribution of X_c given $X_c \equiv b'\ell \pmod a$, where $bb' \equiv 1 \pmod a$. In particular,

$$\mathbb{P}(Y_{(1)} < b) \to \begin{cases} e^{-c}/\mathbb{P}(a \mid X_c), & \ell = 0, \\ 0, & \ell \neq 0, \end{cases} \qquad \text{while} \qquad \mathbb{P}(\xi_{(1)} < b) \to e^{-c}.$$

Hence (i) and (iv) fail for $h = b$, and (v) and (vi) fail for $j = 1$.

(c) If $c_n \to 0$, then the conclusions of Theorem 19.7 hold if and only if $m \to \infty$ and $a \mid m$ for all large n.

Consequently, the conclusions of Theorem 19.7 hold if and only if $m \to \infty$ and every subsequence has a further subsequence along which either $c_n \to \infty$, or $c_n \to 0$ and $a \mid m$.

Example 1.4. Let $w_0 = 1$, $w_1 = 0$ and $w_k = 1$ for $k \geq 2$, so that $\mathbf{Y}$ is uniform on the allocations in which no box holds exactly one ball. Then $\Phi(t) = 1 + t^2/(1-t)$, $\rho = 1$, $\nu = \infty$, $a = 2$, $b = 3$, and (1.1) reads $c_n = (m^3/(8n))^{1/2}(1 + O(\tau_n))$. By Theorem 1.3, the conclusions of Theorem 19.7 hold if $n = o(m^3)$; if $m \to \infty$ and $n/m^3 \to \infty$, they hold along even m and fail along odd m; and if $m^3/(8n) \to c^2$ with $0 < c < \infty$, then for even m

$$\mathbb{P}(N_{[3,\infty)} = k) \to \frac{c^k}{k! \cosh c} \quad (k \text{ even}), \qquad \text{in particular} \qquad \mathbb{P}(Y_{(1)} \leq 2) \to \frac{1}{\cosh c},$$

and for odd m

$$\mathbb{P}(N_{[3,\infty)} = k) \to \frac{c^k}{k! \sinh c} \quad (k \text{ odd}),$$

while $\mathbb{P}(\xi_{(1)} \leq 2) \to e^{-c}$ for both parities.

All of these results come from a comparison of the boxes whose sizes are neither 0 nor a. Let $V = \{0, a\}$. For $x \in \mathbb{Z}^n_{\geq 0}$ let

$$R(x) = (x_i \mathbf{1}\{x_i \notin V\})_{i \leq n},$$

which we call the reduced vector of x, and for $r = R(x)$ let $|r|$ be the number of nonzero coordinates of r and $K(r) = r_1 + \cdots + r_n$. Write $R = R(\boldsymbol{\xi})$ and $R_Y = R(\mathbf{Y})$, and let E_m be the event that $K(R) \equiv m \pmod a$. Since $S_n - K(R)$ is a multiple of a, $E_m = \{S_n \equiv m \pmod a\}$.

Theorem 1.5. *Suppose that $m \to \infty$, $m/n \to 0$ and $\liminf \mathbb{P}(E_m) > 0$. Then $Z(m, n) > 0$ for all large n,*

$$d_{\mathrm{TV}}(\mathcal{L}(R_Y), \mathcal{L}(R \mid E_m)) \to 0, \qquad \text{and} \qquad \mathbb{P}(S_n = m) = \mathbb{P}(E_m)\Big(\frac{a}{2\pi m}\Big)^{1/2}(1 + o(1)).$$

Theorem 1.5 needs no assumption on the span of w or on $\gcd(a, b)$. The vectors themselves are far apart: since $\mathcal{L}(\mathbf{Y}) = \mathcal{L}(\boldsymbol{\xi} \mid S_n = m)$, see Section 3, their total variation distance is $1 - \mathbb{P}(S_n = m)$, which tends to one. Under the assumptions of Theorem 1.1, E_m is the sure event and the reduced vectors are close. Under those of Theorem 1.3(a), $\mathbb{P}(E_m) \to 1/a$, while $K(R_Y) \equiv m \pmod a$ always, so that $d_{\mathrm{TV}}(R_Y, R) \to 1 - 1/a$; only the boxes with more than b balls are close in total variation. When $\gcd(a, b) > 1$ Theorem 1.5 still applies, but the residue of S_n modulo a is then governed by more than one value of $\xi^{(n)}$, and we do not treat this case.

Related work. The random allocations considered here are the case of V. F. Kolchin's generalized allocation scheme [6] in which the ξ_i have a power series distribution; see [7, 10, 2]. For $w_k = 1/k!$ they are the classical allocations of distinguishable balls, for which Poisson limit theorems of this kind are classical [8]. For $a = 1$, the formula for $\mathbb{P}(S_n = m)$ in Theorem 1.5 is contained in a local limit theorem of A. V. Kolchin [3, Theorem 5]; see [1, Lemma 2.1]. Integral and local limit theorems for S_n were obtained in [3] when $w_0 w_1 > 0$, and extended in [4, 5] to weights with $w_0 > 0$, $w_a > 0$ and $w_k = 0$ for $0 < k < a$; see [1, Section 2]. We include a short proof of the formula, since it comes out of the proof of the comparison of reduced vectors. Chuprunov and Fazekas [1, Theorem 2.2] proved a Poisson limit theorem for the number of boxes with exactly r balls among the first K boxes, for bounded r, as $m, n, K \to \infty$ with $m/n \to 0$ and $K\mathbb{P}(\xi^{(n)} = r)$ tending to a positive constant. When $w_0 w_1 > 0$ and $r \geq 2$ this number is a function of R_Y, and their result follows from Theorem 1.5 with $a = 1$.

Two neighbouring questions from [2] are treated in companion papers. Conjecture 19.4 of [2], that $Y_{(1)} = o_p(\log n)$ when $\rho > 0$ and $m/n \to 0$, is proved in [11]. Problem 19.10 of [2] asks whether $d_{\mathrm{TV}}(Y_{(j)}, \xi_{(j)}) \to 0$ for every fixed j under the assumptions of Theorem 19.7; it is answered in [12], which also treats the joint distribution of finitely many $Y_{(j)}$. Theorems 1.1 and 1.3(a) give the corresponding statements when $m/n \to 0$.

Section 2 proves a local limit theorem for the binomial distribution with explicit constants, and Section 3 proves Theorem 1.5. Section 4 derives Janson's conclusions from comparisons of reduced vectors and proves Theorem 1.1 and Proposition 1.2. Section 5 proves Theorem 1.3 and the formulas of Example 1.4, and Section 6 reports exact computations.

2. A local limit theorem for the binomial distribution

For $s > 0$ let

$$\varepsilon(s) = \frac{1}{s} + 4s^{-1/3}e^{-s^{2/3}/5};$$

this function decreases to 0 as $s \to \infty$.

Lemma 2.1. *Let $W \sim \mathrm{Bin}(N, q)$ with $N \geq 1$ and $0 < q < 1$, and let $s^2 = Nq(1 - q)$.*

(a) $\max_j \mathbb{P}(W = j) \leq \sqrt{\pi/8}\,/s$.

(b) If $s \geq 1$, then for every integer j,

$$\left|\sqrt{2\pi}\, s\, \mathbb{P}(W = j) - e^{-(j-Nq)^2/(2s^2)}\right| \leq \varepsilon(s).$$

Proof. Let $v = q(1-q)$, $\chi(t) = 1 - q + qe^{it}$, $\psi(t) = e^{-iqt}\chi(t)$ and $\varphi(t) = \psi(t)^N = \mathbb{E}e^{it(W-Nq)}$. For every integer j, with $x = j - Nq$,

$$\mathbb{P}(W = j) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \varphi(t) e^{-itx}\, dt.$$

Since $|\psi(t)|^2 = 1 - 2v(1-\cos t)$ and $1 - \cos t \geq 2t^2/\pi^2$ for $|t| \leq \pi$,

$$|\varphi(t)| \leq \left(1 - 4vt^2/\pi^2\right)^{N/2} \leq e^{-2s^2t^2/\pi^2}, \qquad |t| \leq \pi.$$

Hence $\mathbb{P}(W = j) \leq \frac{1}{2\pi}\int_{\mathbb{R}} e^{-2s^2t^2/\pi^2} dt = \sqrt{\pi/8}/s$, which is (a).

For (b), note that $\operatorname{Re}\chi(t) \geq 1 - q/2 > 0$ for $|t| \leq \pi/3$, and let $f(t) = -iqt + \log\chi(t)$ there, with the principal branch of the logarithm, so that $\varphi = e^{Nf}$. Then $f(0) = f'(0) = 0$, $f''(t) = -ve^{it}/\chi(t)^2$, so that $f''(0) = -v$, and

$$f'''(t) = -\frac{ive^{it}(1 - q - qe^{it})}{\chi(t)^3}.$$

With $\gamma = \cos t$ we have $|1 - q - qe^{it}|^2 = 1 - 2v(1+\gamma)$ and $|\chi(t)|^2 = 1 - 2v(1-\gamma)$. If $\gamma \geq 1/2$, Bernoulli's inequality $(1-y)^3 \geq 1 - 3y$, with $y = 2v(1-\gamma) \in [0,1]$, gives $|\chi(t)|^6 \geq 1 - 6v(1-\gamma) \geq 1 - 2v(1+\gamma)$. Hence $|f'''(t)| \leq v$ for $|t| \leq \pi/3$, and by Taylor's theorem

$$\left|Nf(t) + \frac{s^2t^2}{2}\right| \leq \frac{s^2|t|^3}{6}, \qquad |t| \leq \frac{\pi}{3}.$$

Let $T = s^{-2/3}$, so that $T \leq 1 < \pi/3$. For $|t| \leq T$ the right side is at most $1/6$, and $|e^z - 1| \leq |z|e^{|z|}$ gives

$$\left|\varphi(t) - e^{-s^2t^2/2}\right| \leq e^{1/6}\, \frac{s^2|t|^3}{6}\, e^{-s^2t^2/2}.$$

Since $e^{-x^2/(2s^2)} = \frac{s}{\sqrt{2\pi}}\int_{\mathbb{R}} e^{-s^2t^2/2} e^{-itx}\, dt$,

$$\left|\sqrt{2\pi}\, s\, \mathbb{P}(W = j) - e^{-x^2/(2s^2)}\right| \leq \frac{s}{\sqrt{2\pi}}\,(I_1 + I_2 + I_3),$$

where

$$I_1 = \int_{|t|\leq T} \left|\varphi(t) - e^{-s^2t^2/2}\right| dt, \qquad I_2 = \int_{T<|t|\leq\pi} |\varphi(t)|\, dt, \qquad I_3 = \int_{|t|>T} e^{-s^2t^2/2}\, dt.$$

Using $\int_{\mathbb{R}} |t|^3 e^{-s^2t^2/2} dt = 4/s^4$ and $\int_T^\infty e^{-\beta t^2} dt \leq e^{-\beta T^2}/(2\beta T)$ for $\beta > 0$, we get

$$I_1 \leq \frac{2e^{1/6}}{3s^2}, \qquad I_2 \leq \frac{\pi^2}{2s^2T} e^{-2s^2T^2/\pi^2} = \frac{\pi^2}{2s^{4/3}} e^{-2s^{2/3}/\pi^2}, \qquad I_3 \leq \frac{2}{s^2T} e^{-s^2T^2/2} = \frac{2}{s^{4/3}} e^{-s^{2/3}/2}.$$

Since $2e^{1/6}/(3\sqrt{2\pi}) < 0.32$, $(\pi^2+4)/(2\sqrt{2\pi}) < 2.8$ and $1/2 > 2/\pi^2 > 1/5$,

$$\frac{s}{\sqrt{2\pi}}(I_1 + I_2 + I_3) \leq \frac{0.32}{s} + 2.8\, s^{-1/3} e^{-2s^{2/3}/\pi^2} \leq \varepsilon(s). \qquad \square$$

3. The reduced vectors

In this section and the next, $\tau = \tau_n$ and $\pi_k = \mathbb{P}(\xi^{(n)} = k) = w_k\tau^k/\Phi(\tau)$; we write

$$p = \mathbb{P}(\xi^{(n)} \notin V), \qquad \kappa = \mathbb{E}\big[\xi^{(n)}; \xi^{(n)} \notin V\big], \qquad \kappa_2 = \mathbb{E}\big[(\xi^{(n)})^2; \xi^{(n)} \notin V\big], \qquad q = \frac{\pi_a}{\pi_0 + \pi_a}.$$

Since $m/n \to 0$, $\tau \to 0$. If $w_k > 0$ and $k \notin V$, then $k > a$. Hence, as $n \to \infty$,

$$\begin{gathered} p \le \kappa \le \kappa_2 = O(\tau^{a+1}), \qquad a\pi_a = \frac{m}{n}(1 + O(\tau)), \qquad n\kappa = O(m\tau), \\ \frac{w_a\tau^a}{2w_0} \le q = O(\tau^a), \qquad s_n^2 := nq(1-q) = \frac{m}{a}(1 + O(\tau)). \end{gathered} \tag{3.1}$$

Indeed, fix $\tau_* \in (0, \rho)$; for $\tau \le \tau_*$ we have $\kappa_2 \le \tau^{a+1} w_0^{-1} \sum_{k>a} k^2 w_k \tau_*^{k-a-1}$. Next, $a\pi_a + \kappa = \mathbb{E}\xi^{(n)} = m/n$, and $\pi_a \ge w_a\tau^a/(2w_0)$ for large n because $\Phi(\tau) \to w_0$; this gives the second and third relations. Finally $\pi_a \le q \le \pi_a/\pi_0 = w_a\tau^a/w_0$, and $nq = n\pi_a/(1-p)$. In particular, $\kappa_2/q = O(\tau)$.

The allocation $\mathbf{Y}$ has the law of $\boldsymbol{\xi}$ conditioned on $S_n = m$: for $y \in \mathcal{B}_{m,n}$, $\mathbb{P}(\boldsymbol{\xi} = y) = w(y)\tau^m/\Phi(\tau)^n$ is proportional to $w(y)$, and summing over $\mathcal{B}_{m,n}$,

$$\mathbb{P}(S_n = m) = \frac{Z(m,n)\tau^m}{\Phi(\tau)^n}.$$

These two statements hold for every $\tau \in (0, \rho)$, with the ξ_i distributed as $(w_k\tau^k/\Phi(\tau))_{k\ge 0}$, and so does the following identity, with q computed from τ. For $N \ge 0$ let $W_N \sim \mathrm{Bin}(N, q)$, and for r in the range of R let

$$g(r) = \mathbb{P}\big(aW_{n-|r|} = m - K(r)\big).$$

Given $R = r$, the ξ_i with $r_i = 0$ are independent, equal to a with probability q and to 0 otherwise. Hence $S_n - K(r)$ is distributed as $aW_{n-|r|}$, so that $\mathbb{P}(S_n = m \mid R = r) = g(r)$, and by Bayes' formula

$$\mathbb{P}(R_Y = r) = \frac{\mathbb{P}(R = r)\, g(r)}{\mathbb{P}(S_n = m)}, \qquad \mathbb{E}g(R) = \mathbb{P}(S_n = m). \tag{3.2}$$

Note that $g(r) = 0$ unless $K(r) \equiv m \pmod a$. In particular, R_Y takes only values r with $K(r) \equiv m \pmod a$.

Proof of Theorem 1.5. Let $s = s_n$ and $E = E_m$, and also write E for the set of values r of R with $K(r) \equiv m \pmod a$. For r in the range of R let $\Lambda(r) = \sqrt{2\pi}\, s\, g(r)$ and

$$D(r) = \frac{m - K(r)}{a} - (n - |r|)q,$$

the distance from the point at which $g(r)$ evaluates the binomial probability to the mean of $W_{n-|r|}$. Since $m = n(a\pi_a + \kappa)$ and $q(1-p) = \pi_a$,

$$D(R) = q\big(|R| - np\big) + \frac{n\kappa - K(R)}{a}.$$

Here $|R| \sim \mathrm{Bin}(n, p)$, and $K(R)$ is a sum of n independent terms with mean κ and second moment κ_2. Hence, by (3.1),

$$\frac{\mathbb{E}|D(R)|}{s} \le \frac{q\sqrt{np} + a^{-1}\sqrt{n\kappa_2}}{\sqrt{nq(1-q)}} = \sqrt{\frac{qp}{1-q}} + \frac{1}{a}\sqrt{\frac{\kappa_2}{q(1-q)}} = O(\tau^{1/2}).$$

Let $r \in E$ with $|r| \le n/2$, and let $s_r^2 = (n-|r|)q(1-q) = s^2(1-|r|/n) \ge s^2/2$. Since $s \to \infty$, for large n we have $s_r \ge 1$, and Lemma 2.1(b), applied to $W_{n-|r|}$ at the point $(m-K(r))/a$, gives

$$\Big|\sqrt{2\pi}\, s_r g(r) - e^{-D(r)^2/(2s_r^2)}\Big| \le \varepsilon(s_r) \le \varepsilon(s/\sqrt{2}).$$

Using $s/s_r \le \sqrt{2}$, $|e^{-y^2/2}-1| \le |y|$ for real y, and $(1-x)^{-1/2}-1 \le 2x$ for $0 \le x \le 1/2$, we obtain

$$\begin{aligned}|\Lambda(r)-1| &\le \frac{s}{s_r}\Big|\sqrt{2\pi}\, s_r g(r) - e^{-D(r)^2/(2s_r^2)}\Big| + \frac{s}{s_r}\Big|e^{-D(r)^2/(2s_r^2)}-1\Big| + \Big|\frac{s}{s_r}-1\Big| \\ &\le \sqrt{2}\,\varepsilon(s/\sqrt{2}) + \frac{2|D(r)|}{s} + \frac{2|r|}{n}.\end{aligned}$$

If $|r| > n/2$ we use $0 \le \Lambda(r) \le \sqrt{2\pi}\, s$. Since $\mathbb{P}(|R| > n/2) \le 2^n p^{n/2} = (4p)^{n/2}$, $p \to 0$ and $s^2 \le n$, it follows that

$$\delta_n := \mathbb{E}\big[|\Lambda(R)-1|\mathbf{1}_E\big] \le \sqrt{2}\,\varepsilon(s/\sqrt{2}) + \frac{2\mathbb{E}|D(R)|}{s} + 2p + \big(\sqrt{2\pi}\, s + 1\big)(4p)^{n/2} \to 0.$$

Since $\Lambda = 0$ off E, (3.2) gives $\mu_n := \mathbb{E}\Lambda(R) = \mathbb{E}[\Lambda(R)\mathbf{1}_E] = \sqrt{2\pi}\, s\, \mathbb{P}(S_n = m)$, and therefore $|\mu_n - \mathbb{P}(E)| \le \delta_n$. As $\liminf \mathbb{P}(E) > 0$, $\mu_n > 0$ for large n, so that $Z(m,n) > 0$; and since $s^2 = (m/a)(1+O(\tau))$,

$$\mathbb{P}(S_n = m) = \frac{\mu_n}{\sqrt{2\pi}\, s} = \mathbb{P}(E)\Big(\frac{a}{2\pi m}\Big)^{1/2}(1+o(1)).$$

Finally, by (3.2), $\mathbb{P}(R_Y = r) = \mathbb{P}(R=r)\Lambda(r)/\mu_n$, while $\mathbb{P}(R = r \mid E) = \mathbb{P}(R=r)\mathbf{1}_E(r)/\mathbb{P}(E)$. Hence

$$d_{\mathrm{TV}}\big(\mathcal{L}(R_Y), \mathcal{L}(R \mid E)\big) = \frac{1}{2}\mathbb{E}\Big[\Big|\frac{\Lambda(R)}{\mu_n} - \frac{1}{\mathbb{P}(E)}\Big|\mathbf{1}_E\Big] \le \frac{1}{2}\Big(\frac{\delta_n}{\mu_n} + \frac{|\mathbb{P}(E)-\mu_n|}{\mu_n}\Big) \le \frac{\delta_n}{\mu_n} \to 0. \qquad \square$$

4. From reduced vectors to Janson's conclusions

For an integer $h_0 \ge 1$ and $x \in \mathbb{Z}^n_{\ge 0}$ let $F_{h_0}(x) = (x_i \mathbf{1}\{x_i \ge h_0\})_{i \le n}$.

Lemma 4.1. *Let $h_0 \ge 1$ be fixed. Suppose that $d_{\mathrm{TV}}(F_{h_0}(\mathbf{Y}), F_{h_0}(\boldsymbol{\xi})) \to 0$ and that $N_{[h_0-1,\infty)} \to \infty$ and $N^{\xi}_{[h_0-1,\infty)} \to \infty$ in probability. Then the conclusions of Theorem 19.7 hold, and $d_{\mathrm{TV}}((Y_{(1)},\dots,Y_{(j)}),(\xi_{(1)},\dots,\xi_{(j)})) \to 0$ for every fixed j.*

Proof. Fix j. For $u \in \mathbb{Z}^n_{\ge 0}$ let $u_{(1)} \ge \cdots \ge u_{(n)}$ be its coordinates in decreasing order, and let $H(u) = (\max(u_{(i)}, h_0-1))_{i \le j}$. If $x \in \mathbb{Z}^n_{\ge 0}$ has at least j coordinates that are at least h_0-1, then $(x_{(1)},\dots,x_{(j)}) = H(F_{h_0}(x))$: for $i \le j$, if $x_{(i)} \ge h_0$ then $F_{h_0}(x)_{(i)} = x_{(i)}$, and if $x_{(i)} = h_0 - 1$ then fewer than i coordinates of $F_{h_0}(x)$ are nonzero, so $F_{h_0}(x)_{(i)} = 0$ and the ith coordinate of $H(F_{h_0}(x))$ is $h_0 - 1$. Hence

$$d_{\mathrm{TV}}\big((Y_{(i)})_{i\le j}, (\xi_{(i)})_{i \le j}\big) \le d_{\mathrm{TV}}\big(F_{h_0}(\mathbf{Y}), F_{h_0}(\boldsymbol{\xi})\big) + \mathbb{P}\big(N_{[h_0-1,\infty)} < j\big) + \mathbb{P}\big(N^{\xi}_{[h_0-1,\infty)} < j\big) \to 0.$$

This proves the statement about $(Y_{(1)},\dots,Y_{(j)})$, and with it (v) and (vi).

For (iv), if $h \ge h_0$ then $N_{[h,\infty)}(x)$ is a function of $F_{h_0}(x)$, so that $\tilde{d}_{\mathrm{K}}(N_{[h,\infty)}, N^{\xi}_{[h,\infty)}) \le d_{\mathrm{TV}}(F_{h_0}(\mathbf{Y}), F_{h_0}(\boldsymbol{\xi}))$. If $h \le h_0 - 1$, then $N_{[h,\infty)} \ge N_{[h_0-1,\infty)}$ and $N^{\xi}_{[h,\infty)} \ge N^{\xi}_{[h_0-1,\infty)}$. Since $\tilde{d}_{\mathrm{K}}(X,\infty) = \sup_x \mathbb{P}(X \le x)/(1+x)$ can only decrease when X is increased,

$$\tilde{d}_{\mathrm{K}}\big(N_{[h,\infty)}, N^{\xi}_{[h,\infty)}\big) \le \tilde{d}_{\mathrm{K}}\big(N_{[h_0-1,\infty)}, \infty\big) + \tilde{d}_{\mathrm{K}}\big(N^{\xi}_{[h_0-1,\infty)}, \infty\big) \to 0$$

by [2, Lemma 19.5]. Both bounds are uniform in h, which gives (iv) for every sequence $h(n)$.

For (i), suppose that $n\mathbb{P}(\xi^{(n)} \geq h) \to \alpha$ along a subsequence, and let $p_h = \mathbb{P}(\xi^{(n)} \geq h)$. If $\alpha < \infty$, then $p_h \to 0$, and Le Cam's inequality $d_{\mathrm{TV}}(\mathrm{Bin}(n, p_h), \mathrm{Po}(np_h)) \leq np_h^2$ [9] shows that $N^{\xi}_{[h,\infty)} \to \mathrm{Po}(\alpha)$ in distribution; if $\alpha = \infty$, then $N^{\xi}_{[h,\infty)} \to \infty$ in probability by Chebyshev's inequality. In both cases $\tilde{d}_{\mathrm{K}}(N^{\xi}_{[h,\infty)}, \mathrm{Po}(\alpha)) \to 0$ by [2, Lemma 19.5], and (iv) together with the triangle inequality gives $\tilde{d}_{\mathrm{K}}(N_{[h,\infty)}, \mathrm{Po}(\alpha)) \to 0$, which is (i). The cases $\alpha = 0$ and $\alpha = \infty$ of (i) give (ii) and (iii). □

Lemma 4.2. *Suppose that $m \to \infty$, $m/n \to 0$ and $\mathbb{P}(E_m) \to 1$. Then $d_{\mathrm{TV}}(R_Y, R) \to 0$, the conclusions of Theorem 19.7 hold, and $d_{\mathrm{TV}}((Y_{(1)}, \dots, Y_{(j)}), (\xi_{(1)}, \dots, \xi_{(j)})) \to 0$ for every fixed j.*

Proof. For any event A with $\mathbb{P}(A) > 0$ and any random element X, $d_{\mathrm{TV}}(\mathcal{L}(X \mid A), \mathcal{L}(X)) \leq 1 - \mathbb{P}(A)$. Hence Theorem 1.5 gives $d_{\mathrm{TV}}(R_Y, R) \to 0$. We apply Lemma 4.1 with $h_0 = a + 1$. Every integer $k \geq a + 1$ lies outside V, so $F_{a+1}(x) = F_{a+1}(R(x))$ and $d_{\mathrm{TV}}(F_{a+1}(\mathbf{Y}), F_{a+1}(\boldsymbol{\xi})) \leq d_{\mathrm{TV}}(R_Y, R) \to 0$. Next, $N^{\xi}_{[a,\infty)}$ is binomial with mean at least $n\pi_a$, which tends to infinity by (3.1), so $N^{\xi}_{[a,\infty)} \to \infty$ in probability. Finally, the boxes of $\mathbf{Y}$ with sizes in V hold $m - K(R_Y)$ balls, so that $N_{[a,\infty)} \geq N_{\{a\}} = (m - K(R_Y))/a$, and by Markov's inequality and (3.1)

$$\mathbb{P}(K(R_Y) > m/2) \leq \mathbb{P}(K(R) > m/2) + d_{\mathrm{TV}}(R_Y, R) \leq \frac{2n\kappa}{m} + d_{\mathrm{TV}}(R_Y, R) \to 0.$$

Hence $N_{[a,\infty)} \geq m/(2a)$ with probability tending to one, and Lemma 4.1 applies. □

Proof of Theorem 1.1. If every k with $w_k > 0$ is a multiple of a, then S_n is a multiple of a, so $Z(m, n) > 0$ forces $a \mid m$, and then E_m is the sure event. If $m \to \infty$, Lemma 4.2 gives all the assertions. If $m \not\to \infty$, then m takes some value m_0 for infinitely many n, and the conclusions fail by Proposition 1.2(a). □

Proof of Proposition 1.2. (a) As $n \to \infty$ in $\mathcal{N}$ we have $\tau \to 0$, and (3.1) gives $n\pi_a = (m_0 - n\kappa)/a$ with $n\kappa = O(m_0\tau) \to 0$, and $np \leq n\kappa$. Hence $n\mathbb{P}(\xi^{(n)} \geq 1) = n\pi_a + np \to m_0/a$, and $\mathbb{P}(\xi_{(1)} = 0) = (1 - \mathbb{P}(\xi^{(n)} \geq 1))^n \to e^{-m_0/a}$. On the other hand, $Y_{(1)} \geq 1$ because $m_0 \geq 1$, and every nonempty box holds at least a balls, so $N_{[1,\infty)} \leq m_0/a$. Thus $\mathbb{P}(N_{[1,\infty)} = 0) = 0$ while $\mathbb{P}(X_{m_0/a} = 0) > 0$, so (i) fails for $h = 1$, and

$$\tilde{d}_{\mathrm{K}}\big(N_{[1,\infty)}, N^{\xi}_{[1,\infty)}\big) \geq \mathbb{P}\big(N^{\xi}_{[1,\infty)} = 0\big) = \mathbb{P}(\xi_{(1)} = 0),$$
$$d_{\mathrm{TV}}(Y_{(1)}, \xi_{(1)}) \geq d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)}) \geq \mathbb{P}(\xi_{(1)} = 0),$$

and these lower bounds tend to $e^{-m_0/a} > 0$.

(b) If $a \nmid m$, then some Y_i is not a multiple of a, and since $w_{Y_i} > 0$, $Y_i \geq b$. Next, $\mathbb{P}(\xi^{(n)} \geq b) = \pi_b + \sum_{k>b} \pi_k$, and $\sum_{k>b} \pi_k = O(\tau^{b+1}) = O(\tau\pi_b)$ as in (3.1); hence $n\mathbb{P}(\xi^{(n)} \geq b) = c_n(1 + O(\tau))$. If $c_n \to 0$, then $\mathbb{P}(\xi_{(1)} \geq b) \leq n\mathbb{P}(\xi^{(n)} \geq b) \to 0$ while $\mathbb{P}(Y_{(1)} \geq b) = 1$. Thus $d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)}) \geq \mathbb{P}(\xi_{(1)} < b) \to 1$, (i) fails for $h = b$ with $\alpha = 0$, (ii) fails for $h = b$, and

$$\tilde{d}_{\mathrm{K}}\big(N_{[b,\infty)}, N^{\xi}_{[b,\infty)}\big) \geq \mathbb{P}\big(N^{\xi}_{[b,\infty)} = 0\big) = \mathbb{P}(\xi_{(1)} < b) \to 1.$$

For the last assertion, $m \equiv b \not\equiv 0 \pmod{a}$, the allocation with one box holding b balls and $\lfloor n^{(b-a)/(2b)} \rfloor$ boxes holding a balls shows that $Z(m, n) > 0$ for large n, and $m \asymp n^{(b-a)/(2b)}$, so that (1.1) gives $c_n \asymp n^{(b-a)/(2a)} n^{1-b/a} = n^{-(b-a)/(2a)}$. □

5. Residues

Lemma 5.1. *Let $a \geq 2$ and b be coprime integers, and let $U_1, \dots, U_N$ be independent, identically distributed, integer valued random variables with $\mathbb{P}(U_1 = 0) \geq \alpha$ and $\mathbb{P}(U_1 = b) \geq \beta$. Then for every integer ℓ,*

$$\Big|\mathbb{P}(U_1 + \cdots + U_N \equiv \ell \pmod a) - \frac{1}{a}\Big| \leq \exp\Big(-\alpha\beta\Big(1 - \cos\frac{2\pi}{a}\Big)N\Big).$$

Proof. For $u \in \{1, \dots, a-1\}$ let $\phi_u = \mathbb{E}e^{2\pi i u U_1/a}$. Then

$$\begin{aligned}|\phi_u|^2 &= 1 - \sum_{x,y} \mathbb{P}(U_1 = x)\mathbb{P}(U_1 = y)\Big(1 - \cos\frac{2\pi u(x-y)}{a}\Big)\\ &\leq 1 - 2\alpha\beta\Big(1 - \cos\frac{2\pi ub}{a}\Big) \leq 1 - 2\alpha\beta\Big(1 - \cos\frac{2\pi}{a}\Big),\end{aligned}$$

since every term of the sum is nonnegative, the terms with $(x, y) = (0, b)$ and $(b, 0)$ contribute at least $2\alpha\beta(1 - \cos(2\pi ub/a))$, and $ub \not\equiv 0 \pmod a$. Hence

$$\Big|\mathbb{P}\Big(\sum_i U_i \equiv \ell \pmod a\Big) - \frac{1}{a}\Big| = \Big|\frac{1}{a}\sum_{u=1}^{a-1} e^{-2\pi i u\ell/a}\phi_u^N\Big| \leq \Big(1 - 2\alpha\beta\Big(1 - \cos\frac{2\pi}{a}\Big)\Big)^{N/2},$$

and $1 - y \leq e^{-y}$ completes the proof. □

Proof of Theorem 1.3. Every $k < b$ with $w_k > 0$ is a multiple of a. Since π_b is of order τ^b and $\mathbb{P}(\xi^{(n)} > b) = O(\tau^{b+1})$ as in (3.1), we have $n\mathbb{P}(\xi^{(n)} > b) = O(c_n\tau)$. As $b > a$, a coordinate of x that is at least b lies outside V; hence $N_{[b,\infty)}(x)$ and $F_{b+1}(x)$ are functions of $R(x)$. In cases (a) and (b), c_n is bounded below, so $m \to \infty$ by (1.1).

(a) Let $J = \{i : \xi_i > b\}$ and $F = F_{b+1}(\boldsymbol{\xi})$. The vector F determines J and the ξ_i with $i \in J$, and given F the ξ_i with $i \notin J$ are independent with distribution $\mathcal{L}(\xi^{(n)} \mid \xi^{(n)} \leq b)$, which gives mass at least π_0 to 0 and at least π_b to b. By Lemma 5.1, on the event $\{|J| \leq n/2\}$,

$$|\mathbb{P}(E_m \mid F) - 1/a| \leq \eta_n := \exp\Big(-\pi_0\Big(1 - \cos\frac{2\pi}{a}\Big)\frac{c_n}{2}\Big) \to 0,$$

and $\mathbb{P}(|J| > n/2) \leq (4\mathbb{P}(\xi^{(n)} > b))^{n/2} \to 0$. Hence $\mathbb{P}(E_m) \to 1/a$ and

$$d_{\mathrm{TV}}(\mathcal{L}(F \mid E_m), \mathcal{L}(F)) = \frac{\mathbb{E}|\mathbb{P}(E_m \mid F) - \mathbb{P}(E_m)|}{2\mathbb{P}(E_m)} \leq \frac{\eta_n + \mathbb{P}(|J| > n/2)}{\mathbb{P}(E_m)} \to 0.$$

By Theorem 1.5, $d_{\mathrm{TV}}(\mathcal{L}(F_{b+1}(\mathbf{Y})), \mathcal{L}(F \mid E_m)) \leq d_{\mathrm{TV}}(\mathcal{L}(R_Y), \mathcal{L}(R \mid E_m)) \to 0$, and therefore $d_{\mathrm{TV}}(F_{b+1}(\mathbf{Y}), F_{b+1}(\boldsymbol{\xi})) \to 0$. Next, $N^{\xi}_{[b,\infty)}$ is binomial with mean at least $c_n \to \infty$, so it tends to infinity in probability, and by Theorem 1.5, for every fixed k,

$$\mathbb{P}(N_{[b,\infty)} \leq k) \leq \mathbb{P}(N^{\xi}_{[b,\infty)} \leq k \mid E_m) + o(1) \leq \frac{\mathbb{P}(N^{\xi}_{[b,\infty)} \leq k)}{\mathbb{P}(E_m)} + o(1) \to 0.$$

Lemma 4.1 with $h_0 = b + 1$ completes the proof of (a).

(b) Let $M = N^{\xi}_{\{b\}}$, a binomial variable with mean c_n, and $L = N^{\xi}_{(b,\infty)}$, so that $\mathbb{P}(L > 0) \leq n\mathbb{P}(\xi^{(n)} > b) = O(c_n\tau) \to 0$. By Le Cam's inequality, $d_{\mathrm{TV}}(M, X_{c_n}) \leq c_n\pi_b \to 0$, so $M \to X_c$ in total variation. On $\{L = 0\}$ the coordinates of R that differ from b are multiples of a, so $K(R) \equiv bM \pmod a$ and $E_m \cap \{L = 0\} = \{M \equiv b'\ell \pmod a\} \cap \{L = 0\}$. Hence, uniformly over sets $A \subseteq \mathbb{Z}_{\geq 0}$,

$$\mathbb{P}(N^{\xi}_{[b,\infty)} \in A,\ E_m) = \mathbb{P}(M \in A,\ M \equiv b'\ell) + o(1) = \mathbb{P}(X_c \in A,\ X_c \equiv b'\ell) + o(1),$$

with congruences modulo a. In particular $\mathbb{P}(E_m) \to \mathbb{P}(X_c \equiv b'\ell \pmod a) > 0$, and $\mathcal{L}(N^{\xi}_{[b,\infty)} \mid E_m)$ converges in total variation to $\mathcal{L}(X_c \mid X_c \equiv b'\ell \pmod a)$. Since $N_{[b,\infty)}$ is a function of R_Y, Theorem 1.5 shows that $N_{[b,\infty)}$ has the same limit. Taking $A = \{0\}$, $\mathbb{P}(Y_{(1)} < b) = \mathbb{P}(N_{[b,\infty)} = 0)$ tends to $e^{-c}/\mathbb{P}(a \mid X_c)$ if $b'\ell \equiv 0$, that is, if $\ell = 0$, and to 0 otherwise. On the other hand $n\mathbb{P}(\xi^{(n)} \ge b) = c_n(1 + O(\tau)) \to c$, so that $N^{\xi}_{[b,\infty)} \to X_c$ in distribution and $\mathbb{P}(\xi_{(1)} < b) \to e^{-c}$. As $a \ge 2$ and $c > 0$, the law of X_c conditioned on a residue class modulo a differs from $\mathrm{Po}(c)$, and $e^{-c}/\mathbb{P}(a \mid X_c) > e^{-c}$. Hence (i) and (iv) fail for $h = b$, and $d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)}) \ge |\mathbb{P}(Y_{(1)} < b) - \mathbb{P}(\xi_{(1)} < b)|$ does not tend to zero.

(c) If $m \not\to \infty$, the conclusions fail by Proposition 1.2(a), and if $a \nmid m$ for infinitely many n, they fail along those n by Proposition 1.2(b). Conversely, suppose that $m \to \infty$ and $a \mid m$. On the event that no ξ_i is at least b, every coordinate of R is a multiple of a, so that $\mathbb{P}(E_m) \ge 1 - n\mathbb{P}(\xi^{(n)} \ge b) = 1 - c_n(1 + O(\tau)) \to 1$, and Lemma 4.2 applies.

For the last assertion, note that (i)–(vi) hold along the whole sequence if and only if every subsequence has a further subsequence along which they hold. If $m \to \infty$ and every subsequence has a further one along which either $c_n \to \infty$, or $c_n \to 0$ and $a \mid m$, then the conclusions hold by (a) and (c). Conversely, suppose that they hold. Then $m \to \infty$ by Proposition 1.2(a). Every subsequence has a further subsequence along which $c_n \to c \in [0, \infty]$ and $m \equiv \ell \pmod a$ for a fixed ℓ; by (b), $c \in \{0, \infty\}$, and if $c = 0$ then $\ell = 0$ by Proposition 1.2(b). □

Proof of the formulas in Example 1.4. Here $\Phi(t) = 1 + \sum_{k\ge2} t^k = 1 + t^2/(1-t)$, so $\rho = 1$, and $\Psi(t) \to \infty$ as $t \uparrow 1$, so $\nu = \infty$. With $a = 2$, $b = 3$ and $w_0 = w_2 = w_3 = 1$, (1.1) gives $c_n = (m^3/(8n))^{1/2}(1 + O(\tau_n))$. If $m^3/(8n) \to c^2$, Theorem 1.3(b) applies with $b' = 1$: for even m, $\ell = 0$ and $\mathbb{P}(X_c \equiv 0 \pmod 2) = e^{-c}\cosh c$; for odd m, $\ell = 1$ and $\mathbb{P}(X_c \equiv 1 \pmod 2) = e^{-c}\sinh c$. The other two cases are Theorem 1.3(a) and (c). For the computations in Section 6, note that $Z(m,n) = \sum_{j\ge1} \binom{n}{j}\binom{m-j-1}{j-1}$, since $\binom{n}{j}$ counts the choices of j nonempty boxes and $\binom{m-j-1}{j-1}$ the compositions of m into j parts that are at least 2, and that for even m the allocations with $Y_{(1)} \le 2$ are those in which $m/2$ boxes hold two balls each, so that $\mathbb{P}(Y_{(1)} \le 2) = \binom{n}{m/2}/Z(m,n)$. □

6. Computations

The script `code/checks.py` in the repository listed under Data availability uses only the Python standard library, and its output is `code/checks.out`. It computes the following quantities exactly, or by floating-point evaluation of exact formulas, and runs in a few seconds.

Lemma 2.1. On a grid of 88 pairs (N, q) with $1 \le N \le 30000$ and $10^{-4} \le q \le 0.99$, the largest value of $s \max_j \mathbb{P}(W = j)/\sqrt{\pi/8}$ is 0.6585, and over the pairs with $s \ge 1$ the largest ratio of the error in Lemma 2.1(b) to $\varepsilon(s)$ is 0.0489.

The identity (3.2). Both identities in (3.2) are verified in rational arithmetic for $w = (2,1,0,3,1,0,2)$ with $(m,n,\tau) = (7,4,2/5)$, where $a = 1$, and for $w = (1,0,2,1,3,0,1,2)$ with $(m,n,\tau) = (7,4,1/3)$ and $(8,4,3/7)$, where $a = 2$. Replacing q by π_a breaks both identities.

Uniform weights. For $w_k = 1$ for all k, $\mathbf{Y}$ is uniform on $\mathcal{B}_{m,n}$, $a = 1$, and $\xi^{(n)}$ is geometric with $\mathbb{P}(\xi^{(n)} = k) = (1-t)t^k$, where $t = \tau_n = m/(m+n)$. By (3.2) the ratio $\mathbb{P}(R_Y = r)/\mathbb{P}(R = r)$ depends on r only through $(|r|, K(r))$, and $d_{\mathrm{TV}}(R_Y, R)$ can be computed exactly from the joint law of $|R|$ and $K(R)$. Table 1 shows the result and the ratio $\sqrt{2\pi m}\,\mathbb{P}(S_n = m)$, which tends to 1 by Theorem 1.5. Along $n = m^2$ the distance lies between $0.83\,m/n$ and $0.92\,m/n$. For fixed $m/n = 0.1$ the distance stays near 0.09, and the ratio tends to $1.1^{-1/2} = 0.9535$ rather than to 1, because then the variance of S_n is $1.1m$. For the largest box, $d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)})$ equals 0.0301, 0.0115, 0.0036 and 0.0012 for $m = \sqrt{n} = 10$, 30, 100 and 300. For $m = 1$ it equals

$\mathbb{P}(\xi_{(1)} = 0) = (1 - 1/(n+1))^n$, which is 0.3855, 0.3697, 0.36790 and 0.367880 for $n = 10$, 10^2, 10^4 and 10^6 and tends to $e^{-1} = 0.367879$, as in Proposition 1.2(a).

TABLE 1. Weights $w_k = 1$: the distance between the reduced vectors and the local limit ratio.

m	n	m/n	$d_{\mathrm{TV}}(R_Y, R)$	$d_{\mathrm{TV}}(R_Y, R)\, n/m$	$\sqrt{2\pi m}\,\mathbb{P}(S_n = m)$
20	400	0.05	0.041604	0.8321	0.971833
50	2500	0.02	0.017695	0.8847	0.988498
100	10000	0.01	0.009022	0.9022	0.994208
200	40000	0.005	0.004555	0.9110	0.997094
400	160000	0.0025	0.002288	0.9153	0.998544
800	640000	0.00125	0.001147	0.9175	0.999271
100	1000	0.1	0.089278	0.8928	0.952661
200	2000	0.1	0.089667	0.8967	0.953062
400	4000	0.1	0.089825	0.8982	0.953262
800	8000	0.1	0.089912	0.8991	0.953362

Example 1.4. The laws of $Y_{(1)}$ and of $N_{[3,\infty)}$ are computed from exact counts of allocations. For $n = m^2$, where $c_n \to \infty$, $d_{\mathrm{K}}(Y_{(1)}, \xi_{(1)})$ equals 0.156, 0.084, 0.033 and 0.0086 for $m = 20$, 40, 80 and 160, and 0.258, 0.137, 0.055 and 0.015 for $m = 21$, 41, 81 and 161. For $n = m^4$, where $c_n \to 0$, it equals 0.098, 0.072, 0.053 and 0.038 for $m = 10$, 20, 40 and 80, and 0.901, 0.926, 0.946 and 0.962 for $m = 11$, 21, 41 and 81. Table 2 treats $n = \mathrm{round}(m^3/8)$, where $c_n \to 1$; the limits for $c = 1$ are $1/\cosh 1 = 0.648054$ for even m and $k = 0$, and $1/\sinh 1 = 0.850918$ for odd m and $k = 1$, while $\mathbb{P}(\xi_{(1)} \le 2) \to e^{-1} = 0.367879$. Finally, the ratio $\sqrt{\pi m}\,\mathbb{P}(S_n = m)/\mathbb{P}(E_m)$ of Theorem 1.5, with $\mathbb{P}(E_m) = (1 + (-1)^m(\Phi(-\tau_n)/\Phi(\tau_n))^n)/2$, equals 0.990 for $n = m^2$ and $m = 640$, 641; 0.994 and 1.0008 for $n = \mathrm{round}(m^3/8)$ and $m = 320$, 321; and 0.998 for $n = m^4$ and $m = 80$. For odd m and $n = m^4$, $\mathbb{P}(E_m) \to 0$ and Theorem 1.5 does not apply.

TABLE 2. Example 1.4 with $n = \mathrm{round}(m^3/8)$. Here $p_N = \mathbb{P}(N_{[3,\infty)} = k)$ and $p_X = \mathbb{P}(X_{c_n} = k \mid X_{c_n} \equiv m \pmod 2)$, with $k = 0$ for even m and $k = 1$ for odd m; the last two columns are the total variation distances from the law of $N_{[3,\infty)}$ to the conditioned law of X_{c_n} and to $\mathrm{Po}(c_n)$.

m	n	c_n	p_N	p_X	conditioned	$\mathrm{Po}(c_n)$
20	1000	0.8046	0.673407	0.745415	0.080814	0.331224
40	8000	0.8954	0.661004	0.700084	0.045949	0.375237
80	64000	0.9458	0.654613	0.674936	0.024634	0.401993
160	512000	0.9724	0.651356	0.661714	0.012784	0.416621
21	1158	0.8126	0.877126	0.897866	0.056893	0.542524
41	8615	0.8978	0.862768	0.877301	0.037152	0.545932
81	66430	0.9464	0.856480	0.864959	0.021432	0.553893
161	521660	0.9726	0.853601	0.858168	0.011555	0.559930

These computations test the identities and inequalities used in the proofs; they are not part of them.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [13] and working together inside the Brain framework [15], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [14], the Math Brain repository [13] and the Brain framework repository [15].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Problem 19.9 of [2] was among the open problems in probability on which the project's agents worked in September 2026. On 13 September 2026 Claude Fable 5.1 (Anthropic) proved a version of Theorem 1.1 for weights with $w_1 > 0$ and of Proposition 1.2 for weights of span one, with the weights of Example 1.4 as the example. On 13 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found Theorem 1.5 and the proofs of the results in their present form, including Theorem 1.3 and the limits in Example 1.4, searched the literature, and wrote the computer checks and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/sparse-allocation-poisson-limits`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [13].

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